

**GC University Lahore**

**Examination: Final, 2019**

**Semester: V**

**Course: Database Systems**

**Course Code: CS-2014**

**Allowed Time: 25 Minutes**

**Total Marks: 14**

**PART A: Objective Type Questions**

**Note:** Encircle the right answer. Erasing, cutting or overwriting shall bear "No Marks".

**1. Usually designing and ER diagram is \_\_\_\_\_ process.**

- ☐ a) A Sequential
- ☐ b) Random
- ☐ c) An Iterative
- ☐ d) Probabilistic

**2. The type of relationship is abnormal in Relational Database.**

- ☐ a) One-to-one relationship
- ☐ b) One-to-many relationship
- ☐ c) Many-to-many relationship
- ☐ d) Unary relationship

**3. In a relation  $R(A, B, C)$ , A is primary key and C is unique key, C can be \_\_\_\_\_**

- ☐ a) Primary key
- ☐ b) NOT NULL
- ☐ c) Redundant
- ☐ d) Empty

**4. Alter command is part of \_\_\_\_\_**

- ☐ a) Data Manipulation Language
- ☐ b) Data Control Language

- c) Data Retrieval Language
- d) Data Definition Language

5. **For each attribute of relation R there is a permitted values called \_\_\_\_\_ of that attribute.**

- a) Range
- b) Co-domain
- c) Domain
- d) Type

6. **Primary key in a relation is conceptually referred to:**

- a) Entity integrity
- b) Referential integrity
- c) Domain integrity
- d) Partial integrity

7. **A \_\_\_\_\_ represents the number of entity instances to which another entity instances can be associated.**

- a) Data Dictionary
- b) Table
- c) Cardinality
- d) schema

8. **A relational database includes**

- a) Relations
- b) Tuples
- c) Keys
- d) All of the above

9. **An attribute that gets its value from computation of other attributes is \_\_\_\_\_ attribute.**

- a) Multivalued
- b) Composite
- c) Derived

- d) Simple

**10. A domain is atomic if elements of domain are considered to be \_\_\_\_\_ units.**

- a) Different
- b) Composite
- c) Divisible
- d) Indivisible

**11. Only one primary key constraint exists in a relation but multiple \_\_\_\_\_ constraints can exist.**

- a) Unique
- b) Check
- c) Foreign Key
- d) All of the above

**12. Which command makes data in database changes permanent:**

- a) Commit
- b) Alter
- c) Delete
- d) Update

## **PART B: Subjective Questions**

### **Q No. 1.**

At a Movie Rental Store each movie is identified by a movie number and also has a title and director of the movie (if associated is identified ID and Director Name and the Studio (Fox Studio, Evernote Studios) that produced the movie, a studio is located in a city as a specific address and may have more than one phone numbers. There are many actors who play characters in the movies, an actor can have several actor id and each actor can play in many movies. A video store holds multiple copies of the same movie, and the store differentiates copies with a movie copy number Each movie has a type (HD, Blu-Ray, DVD) and rent of the movies varies according to movie type. A customer gets the movie on rent after getting registered with the Movie Shop. A customer is registered on the referral of another customer.

- **a. Draw an ER diagram for this system. (10)**
- **b. Resolve many-to-many or unary relationships in ER Diagram. (3)**

**Q No. 2.**

Given a table:

| C_ID | Client Name     | S_ID | S_Name    | C_P   | C_Idc | Desig             | Salary  | D_ID | DN ame   | D_Tel      |
|------|-----------------|------|-----------|-------|-------|-------------------|---------|------|----------|------------|
| C 1  | HD-Broad band   | 51   | S. Asi m  | 50    | 12 3  | Lect urer         | 50,00 0 | D 1  | CS       | 998, 919   |
|      |                 | 52   | S. Yasi r | 20 0  | 12 3  | Cler k            | 25,00 0 | D 1  | CS       | 998, 919   |
|      |                 | 53   | M. Waji d | 20 0  | 12 4  | Cler k            | 25,00 0 | D 2  | Eng lish | 998, 912   |
|      |                 | 54   | M. Ali    | 20 0  | 12 7  | Cler k            | 35,00 0 | D 3  | Mat h    | +40 1,22 0 |
| C 2  | PTCL-Broad band | 52   | S. Yasi r | 10 0  | 13 5  | Assi stan t Prof. | 80,00 0 | D 1  | Phy sics | 992, 201   |
|      |                 | 51   | S. Asi m  | 12 00 | 13 6  | Assi stan t Prof. | 85,00 0 | D 1  | Phy sics | 313 12     |
|      |                 | 55   | Fou zia   | 90 0  | 13 4  | Lect urer         | 50,00 0 | D 1  | Eng lish | 992, 201   |

| C_ID | Client Name | S_ID | S_Name          | C_P      | C_Idc   | Desig     | Salary         | D_ID   | DN ame      | D_Tel    |
|------|-------------|------|-----------------|----------|---------|-----------|----------------|--------|-------------|----------|
|      |             | 56   | M. Hurr<br>aira | 10<br>00 | 13<br>1 | Cler<br>k | 35,<br>00<br>0 | D<br>1 | Eng<br>lish | 512<br>8 |
|      |             | 57   | M.<br>Ali       | 20<br>0  | 12<br>7 | Cler<br>k | 35,<br>00<br>0 | D<br>3 | Mat<br>h    | 412<br>0 |

**Keys Description:**

- C\_ID: Client Identification Number
- SID: Staff/Person Record ID
- S\_Name: Staff Name
- C\_P: Charge Per Hour
- Desig: Designation
- D\_ID: Department Number
- DName: Department Name
- D\_Tel: Dept Phone Number
- CNIC: Computerized National ID Number
- j) Find out all the functional dependencies (FDs). **(2)**
- ii) Normalized Data into 1NF, 2NF, BCNF, 4NF. **(10)**

**Q No. 4.**

- a) What is a foreign key, Differentiate between Candidate Key and Foreign Key. **(4)**
- b) Mention the detail of Insertion, Modification and Deletion anomalies list. **(6)**
- c) Without mentioning detail, only write down all six phases of Database Life Cycle in appropriate sequence. **(2)**

**Q No. 5.**

- a) During Database Life Cycle what kind of measure are done in, write short answer: **(3)**

- i. Adaptive Maintenance
  - ii. Corrective Maintenance
  - iii. Preventive Maintenance
- **b)** What are concurrent transactions? What is the core objective of concurrency control? Explain lost update in absence of concurrency control. **(5)**
- **c)** What are transaction logs and what role they play in recovery of transaction. **(4)**

**Q No. 6.**

- **a)** What is Data Warehouse? What are tasks performed in Extraction, Transformation Loading? **(6)**
- **b)** What are wait/die and wait/wound schemes? **(6)**